

Ibtesam Hussain

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PROFILE

An Engineer who is skilled in **machine learning, deep learning, and model development/deployment**. Ready to integrate reliable AI models into software systems.

EXPERIENCE

Aykays Startup — AI Software Engineer Intern (Onsite, Karachi, Pakistan)

July 2025 — Sept 2025

- Implemented and fine-tuned **LLMs (GPT-4.1, Mistral 7B)** via LangChain and OpenRouter API, improving understanding and response accuracy.
- Built Flask-based **ML inference APIs** for fine-tuned LLM models and integrated them into deployed AI web apps.
- Delivered **four milestone projects** including an **AI Portfolio Generator, Q&A Forum, Product Catalog, and Booking System** using React, Firebase, Node.js, and MongoDB.
- Collaborated via **Slack, Asana**, and **Git/GitHub** for project management, reviews, and version control.

FAST NUCES — Final Year Project Lead (Hybrid, Karachi, Pakistan)

Sept 2025 — May 2026

- Led a **3-member research team**, delivering a **novel hybrid monocular depth estimation system** for **clinical wound imaging**.
- **Designed a two-pipeline system** combining Gemini AI-synthesized stereo viewpoints with DA3 Giant monocular depth model, fused via Laplacian pyramid technique.
- Achieved **43.1%** of image pairs reaching **calibrated stereo range** with best **reprojection errors of 5.26 gray levels** across the dataset.

SKILLS

LLM & GenAI: LangChain, OpenAI API, OpenRouter, Prompt Engineering, HuggingFace Agentic AI, n8n, CrewAI

Programming Languages: Python, SQL, JavaScript.

DS & ML: Tensor Flow, Keras, Pandas, Pytorch, Scikit-learn, NumPy, OpenCV, Deep Learning (CNNs, RNNs, LSTM), Transfer Learning, YOLO, EDA, Data preprocessing, Feature Engineering, MLflow and TensorBoard.

Tools & Deploy: Model Inference APIs, RESTful APIs, Google Colab, Jupyter Notebook, Visual Studio Code, Slack, Jira, Git/Github.

PROJECTS

Handwritten Digits Recognizer — (Deep Learning, AI)

- Build and trained a CNN-based handwritten digits recognizer using Keras/Tensorflow.
- Performed **data preprocessing, EDA and normalization** on **70,000** handwritten digits images.
- Achieved **97.6% accuracy** on the MNIST dataset (**trained on 101,770 parameters**).
- Deployed inference model via **Flask** for **real-time predictions** on a WebApp.
- Trained **60,000 images** and validated on **10,000 samples**, ensuring strong generalization to unseen handwriting styles.
- Reduced model inference time to **<100ms per prediction**, enabling smooth user interaction through the browser.

YogaPose - Caption Generator — (Computer Vision, Sequence Model)

- Contributed to a real-time yoga-pose-detection system with **90%+ classification accuracy using Tensor Flow and PoseNet**.
- Trained and optimized an **LSTM-based caption generator**, improving caption relevance by **30%** through dataset curation and hyper-parameter tuning.
- Integrated webcam-based OpenCV input, achieving **<200ms latency in real-time video inference** for smooth user interaction.
- Reduced false caption generation by **25% by refining pose thresholds and improving model confidence filtering**.
- Contributed to **10+ code commits**, addressed multiple GitHub issues, and enhanced documentation for onboarding and usability.

ConsciousDay Agent — (AI WebApp, Streamlit, LLM Integration)

- Integrated **DeepSeek R1 LLM** for personalized productivity reflection and journaling.
- Implemented journal management and history-context features achieving **~75% accurate memory retention per user**.
- Integrated performance tracking to visualize user productivity trends and previous AI recommendations.
- Deployed via **Streamlit** with persistent user data handling for improved retention

EDUCATION

FAST NUCES, Karachi, Pakistan

Bachelor of Science in Artificial Intelligence

(Expected June 2026)